A Novel Standardized Optimal Synthesized Drought Index (SOSDI) for Agro-Drought Monitoring Using Multisource Remote Sensing Data

Shouming Feng, Shuhe Zhao, *Member, IEEE*, Xinyi Huang, Xinya Guo, Jinlong Fan, and Zhenke Zhang

***Abstract*—Drought has been putting enormous pressure on agriculture and food security, especially for African countries with inadequate monitoring. However, there is still a lack of standardization in the synthesized drought index, which could better monitor drought than normalized indices, as well as insufficient attention to the geographical backgrounds of weights. We aim to develop a standardized optimal synthesized drought index (SOSDI) that will improve drought monitoring and reflect dominant factors. The standardization transforms the input variable distribution functions, with the constraint optimization to compute the weight. Weight matrices are suggested to examine the geospatial heterogeneity of the input variable. The assessment of SOSDI from several dimensions was implemented. SOSDI was applied to monitor the periodic drought in East Africa, where rain-fed agriculture was susceptible to seasonal alternation. The results confirmed that standardization was more effective than normalization in the synthesized drought index, and the average correlation between SOSDI with the existing indices, soil moisture, and crop yield reached about 0.6, 0.7, and 0.7, respectively. The cyclical drought is well captured on the seasonal and annual scales, mostly during the long-dry season. SOSDI could capture drought evolution and had good resistance to disturbances. Overall, SOSDI demonstrated a strong capability in drought monitoring and enhanced the drought expression derived from vegetation and temperature more effectively, rather than just precipitation.**

***Index Terms*—Agricultural drought, drought monitoring, standardization, surface temperature, weight matrix.**

I. Introduction

DROUGHT is one of the most frequent disasters, affecting the natural environment and the development of human society. It is mainly manifested as water shortage caused by insufficient precipitation [1], [2] and exerts stress on terrestrial ecosystems directly [3]. The negative impacts of drought will reduce the resistance and recovery ability of ecosystems [4] by causing plant water deficit [5] and increasing forest mortality and wildfire [6]. Severe drought decreases crop yield [7] and brings great challenges to global agriculture and food security [8]. For instance, the famine and food crisis caused by the East African drought during 2010–2011 affected more than nine million people [9], [10]. Therefore, effective monitoring of drought is critical, especially as climate change will significantly alter drought characteristics [11].

Drought is dynamic and is divided into meteorological, agricultural, and hydrological socioeconomic types [12]. Ecological drought has been proposed [13] recently. Drought indices from information on the Earth's surface [14] realize agricultural drought monitoring [15], especially in Africa, where droughts are frequent but data are scarce. Univariate drought index only considers a single variable. Multivariate drought index combines two or more variables, such as scPDSI [16] and standardized precipitation evapotranspiration index (SPEI) [17]. The synthesized drought indices are continuously developed and achieve more accurate monitoring through the combis greater than zeroination of several indices.

There remain significant deficiencies in the synthesized drought indices since only normalized methods have been applied [18], [19], [20], [21], ignoring the application of standardization. It is widely acknowledged that standardized indices performed better than normalized indices in univariate drought indices [17], [22], such as the standardized precipitation index (SPI) is better than the precipitation condition index (PCI); however, a standardized synthesized drought index has not been developed, which differs from normalized indices. This research gap not only hinders the effectiveness of indices but also limits the potential integration with broader modeling frameworks. It is necessary to build a standardized synthesized drought index and confirm its better performance, as standardization is more closely related to the occurrence of the basic elements of drought.

The weights determine the combination of several indices and are crucial for the synthesized drought index. The empirical weights are subjective and ignore the spatial variation. Weights based on principal component analysis [22], [23], joint distribution [24], [25], and machine learning [18], [26] have no access to comparison of variables, and an inflexible process limits their application. By contrast, the constrained

Received 31 March 2025; revised 7 July 2025; accepted 7 September 2025. Date of publication 10 September 2025; date of current version 18 September 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42271321 and Grant 42330106, in part by Nanjing University Integrated Research Platform of the Ministry of Education—Top Talents Program under Grant 2024300472, and in part by the National Key Research and Development Program of China under Grant 2018YFE0105900. *(Corresponding authors: Shuhe Zhao; Jinlong Fan.)*

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Digital Object Identifier 10.1109/TGRS.2025.3608277

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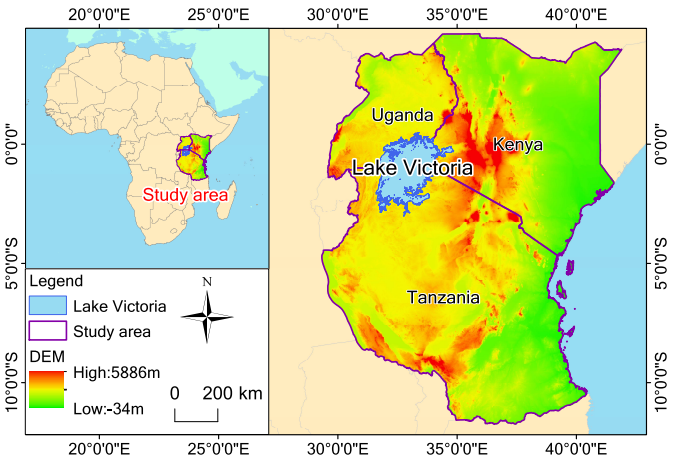


Fig. 1. Location and elevation of the study area in East Africa, including Kenya, Tanzania, and Uganda.

optimization method [20], [21] is more flexible and overcomes the limitations of subjectivity and thus was utilized in the study. However, few studies considered the features of weights over geographic backgrounds [27], and this limits the comparison and potential to assess how inputs take effect. It is necessary to develop spatial weights for better interpretability.

This study aims to develop a standardized optimal synthesized drought index (SOSDI), trying to completely optimize drought monitoring models from normalization to standardization. The role of standardization achieves objective measurement of drought and cross regional comparison by converting data into a unified dimensionless distribution, which is suitable for the distribution of the input through probability conversion. SOSDI fills the gap in the absence of a standardized synthesized drought index. The proposed spatial weight matrix based on constraint optimization enhances our understanding of drought and quantifies the roles of input variables, which are ignored before. The performance of SOSDI was evaluated from multiple dimensions to confirm its effectiveness.

II. Materials and Data Processing

A. Study Area

East Africa is the focus of drought research and crucial for agriculture (Fig. 1). The study area is typical for constructing and analyzing the drought index, covering an area of about 1.77 million km^2. This region is heavily dependent on rain-fed agriculture, which forms the backbone of the three countries' national economies. Rain-fed agriculture is highly sensitive to drought, especially in northeastern Tanzania, eastern Kenya, and northern Uganda, where crop productivity is determined by seasonal rainfall [28]. Three globally important agricultural heritage systems identified by the United Nations in sub-Saharan Africa are all located here [29].

Due to the seasonal alternation in atmospheric circulation and impacts from the Indian Ocean [30], as well as the cyclical effects of atmospheric thermal conditions and water budget balance [31], the precipitation gradient over months is obvious from 50 to 2400 mm. The climate types are complex here and suitable to assess the suitability of SOSDI, including tropical, arid, and temperate climate zones. This region presents bimodal precipitation, concentrated in two rainy seasons: the long-wet season (March to May) and short-wet season (October to December). The long-dry season (June to September) and short-dry season (January to February) are separated by two rainy seasons; 70% of the areas are interleaved with woodland and grassland, with increased cropland and decreased forest. Lake Victoria, the world's second-largest freshwater lake, feeds more than 30 million people and nourishes agriculture.

B. SOSDI Input Data

According to the World Meteorological Organization [32], at least 30 years of records are required to calculate standardized indices. Here, 40 years of precipitation, land surface temperature, and normalized difference vegetation index (NDVI) were used as the inputs, and two widely used drought indices as the objective function.

1) Climate hazards group infrared precipitation with stations (CHIRPSs) monthly precipitation product integrates the long-term observation of measured data and products [33] and is suitable for African studies, where observational data are scarce. The high accuracy and effectiveness of CHIRPS in East Africa have been verified [34].
2) The land surface temperature was derived from ECMWF Reanalysis v5 (ERA5) datasets and provided by the ECMWFs. It has a stronger consistency derived from reanalysis [35], which is realized through the numerical integration of ECMWF's surface model, considering water resources and energy cycles on the surface.
3) Long time series (1982–2021) of NDVI was constructed to meet the requirement of standardization. Global inventory monitoring and modeling study (GIMMS) NDVI is the earliest documented NDVI and could be extended to other products [36]. It is derived from the advanced very high-resolution radiometer (AVHRR) sensor and is reconstructed by the empirical mode decomposition. Data uncertainty is reduced by parameters through Bayesian transformations [37, p. 20]. Moderate resolution imaging spectroradiometer (MODIS) NDVI is a supplement to GIMMS NDVI, which is obtained from the Terra environmental satellite. The latest version of MOD13C2 [38] was selected, and all cloud coverage areas were replaced with historical data.
4) Self-calibrating palmer drought severity index (sc-PDSI) is well applied and more comparable to the PDSI [39] by automatically correcting parameters, such as climate weight factors. It is based on climatic research unit (CRU) data [40], which is calculated according to the precipitation, temperature, and the physical and chemical properties of soil.
5) SPEI is the improvement and optimization of SPI [17]. SPEI reflects the impact of evapotranspiration in drought through the process of climate water balance, which has

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the advantage of multiscales. The SPEI database product [41] with a time scale of 1–6 months was selected to verify the ability of SOSDI to characterize drought over multiple time scales.

C. SOSDI Validation and Analysis Data

Soil moisture and crop yield data were used to validate the performance of SOSDI. Soil moisture is an important parameter for agricultural drought; therefore, it is used as validation data rather than input data to better validate the characterization of agricultural drought. The detection based on remote sensing has a good monitoring effect on Earth's surface soil moisture [42] to reflect drought evolution.

1) The climate change initiative (CCI) surface soil moisture product provided by the European Space Agency integrates multisource remote sensing data [43] and was retrieved from radiation-corrected backscatter data or luminance temperature to generate a uniform, long-period dataset. CCI is considered to be one of the best-performing soil moisture products [44]. The combined product was selected, which coordinated active and passive datasets and performed better in both dense and sparsely vegetated areas.
2) The climate prediction center (CPC) soil moisture product of the US National Weather Service was produced by the global precipitation observation and land surface process model [45]. The model-driven soil moisture dataset CPC provides simulation results dating back to 1948.
3) The National Aeronautics and Space Administration (NASA)'s Global Land Data Assimilation System (GLDAS) integrates the amount of observational data [46] by driving models of different land surface processes. GLDAS_NOAH025_M 2.1 product driven by Noah land surface model (NOAH-LSM) was selected for verification [47].
4) The World Bank provides lots of development indicators and has the most well-documented crop yield information [48]. Two indicators were derived from the statistics of the Food and Agriculture Organization of the United Nations, including cereal yield and crop production index.
5) GlobeLand30 land cover was extracted and generated from multitemporal Landsat data and environmental satellite images [49], [50], with a spatial resolution of 30 m. It reached an overall accuracy of more than 80%. GlobeLand30 combined pixel- and object-based classification integration methods with prior knowledge to divide land use into ten first-level types. GlobeLand30 presented the most accurate classification of land surface, especially cultivated land in the study area.
6) To analyze the influence of climate zones on the weight matrix of SOSDI, the improved Koppen–Geiger climate zoning data were introduced [51], which combined precipitation and temperature data from multiple independent sources to improve the Koppen climate classification map.

D. Data Processing

Use the resampling method to ensure that all data have the same resolution of 0.5°. The maximum value composite method was used to merge half-month GIMMS NDVI to obtain monthly NDVI data [52], so as to reduce the influence of cloud and water vapor [53]. NDVI was calculated after resampling the reflectance of near-infrared (NIR) and red wavelengths (red) data to 0.5° since the calculation was a nonlinear transformation. NDVI is a ratio expressed as (NIR − Red)/(NIR + Red) [54]. Using the moving average method to fill the missing data of the MODIS NDVI. Taking one pixel as the center, the average of the four adjacent years of the center was used to fill in the vacancy value. The missing value of adjacent years was ignored. The formula was as follows:

$$\mathrm{NDVI}_{i,y} = \frac{\mathrm{NDVI}_{i,y-2} + \mathrm{NDVI}_{i,y-1} + \mathrm{NDVI}_{i,y+1} + \mathrm{NDVI}_{i,y+2}}{4} \quad (1)$$

where $\mathrm{NDVI}_{i,y}$ represents the filling results at the i location in the year y, and the rest represents the data of adjacent years.

Due to the sensors' differences, AVHRR is strongly affected by water vapor in the NIR band, while MODIS avoids the spectral range where water vapor absorption is strong. MODIS NDVI is considered to be of high accuracy [55] and is constantly updated. The fitted GIMMS NDVI during 1982–2000 and MODIS NDVI during 2000–2021 were combined. Based on the overlapping range during 2000–2015, GIMMS NDVI was fitted to MODIS NDVI using the least-squares method at the pixel level [56]. R, PBIAS, and RMSE were utilized to evaluate the suitability of the fitting.

The area of GlobeLand30, which was farmland in the three time periods, was extracted as an invariant area for evaluations. Therefore, crop yield and crop production index were valid indicators, rather than crop production, which represented the whole farmland. The crop production index represented the agricultural output relative to the base period from 2014 to 2016; thus, it was the sum of the three countries. The cereal yield of the whole region was expressed as

$$\mathrm{Crop_all}_n = \frac{\sum_{i=1}^{3} \left(\mathrm{Crop}_{i,n} \times S_i\right)}{\sum_{i=1}^{3} S_i} \quad (2)$$

where $\mathrm{Crop_all}_n$ represents the cereal yield of the whole area in the year n, $\mathrm{Crop}_{i,n}$ represents the cereal yield of i country in the year n, S_i represents an invariant area of i country, and the value of i indicates Kenya, Tanzania, and Uganda, respectively. To exclude the influence of human activities, technological progress, and climate change on crops, detrending was carried out on the annual crop indicators [57] to better reflect the deviation. Detrending was based on the least squares method, and parameters were estimated by the following equations:

$$a = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \quad (3)$$

$$b = \hat{y} - a\hat{x} \quad (4)$$

$$\mathrm{crop}_i = y_i - (ax_i + b) \quad (5)$$

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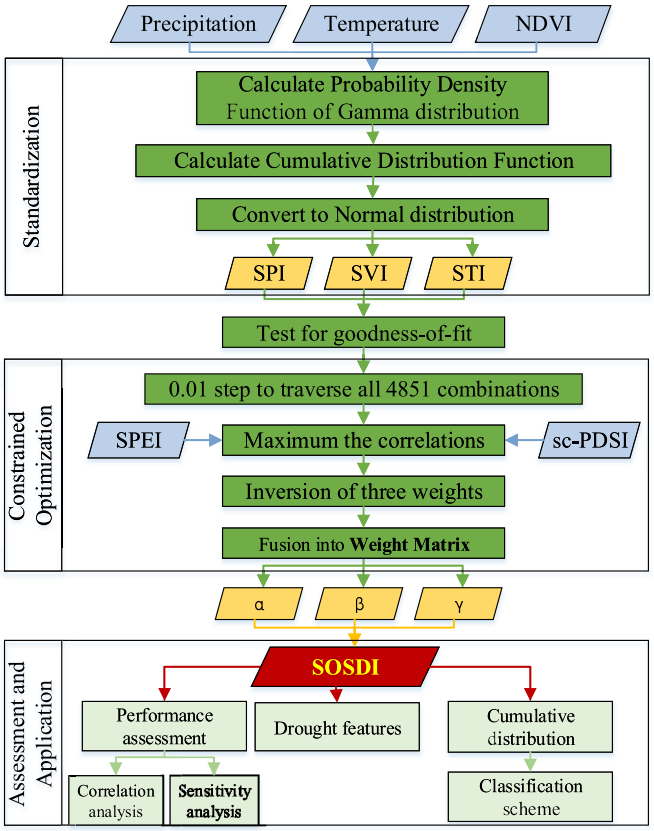


Fig. 2. Framework of the development, evaluation, and application of SOSDI.

where $\hat{y}$ is the variation, $\hat{x}$ is the year, a is the estimated linear coefficient, and b is the residual term. n is the number of observations, x_i is the independent variable, and y_i is the dependent variable. $\bar{x}$ and $\bar{y}$ represent the mean values of x_i and y_i, respectively. crop_i is the detrended crop indicator, and other drought indices were also detrended.

Precipitation data at 0.05°, NDVI, and temperature data at 0.1° were standardized during 1982–2022. The calculated SPI, SVI, and STI were resampled to 0.5° using the nearest neighbor method, consistent with the objective functions of SPEI and scPDSI. All data ensured the same spatial resolution of 0.5°, with a monthly scale during 2000–2021.

III. Methodology of SOSDI

Fig. 2 illustrates the development of SOSDI. The univariate drought indices SPI, standardized vegetation index (SVI), and standardized temperature index (STI) were calculated as an input through standardization. Then, the constrained optimization method was utilized to compute the optimal weight matrix with SPEI and scPDSI as objective functions and synthesized into SOSDI. Performance assessment was conducted according to the correlation analysis. SOSDI was divided into 11 categories based on the cumulative distribution of values and represented drought of varying intensity. At last, SOSDI was utilized to monitor spatiotemporal features in the East Africa.

A. Calculation of Univariate Drought Index

The normalized indices PCI, VCI, and TCI based on precipitation, NDVI, and temperature are calculated as follows:

$$\text{CI} = \frac{\text{CI}_{i,t} - \text{CI}_{i,min}}{\text{CI}_{i,max} - \text{CI}_{i,min}} \tag{6}$$

where CI represents one of PCI, VCI, or TCI. $\text{CI}_{i,t}$ represents the value of pixel i in the period t; $\text{CI}_{i,min}$ and $\text{CI}_{i,\max}$ represent the minimum and maximum value of pixel i in the time series, respectively. Normalization is achieved by scaling the data to 0–1 with only proportional value changes. Standardization is the extraction of features from fitted functions by changing the data distribution [17]. Then, we calculate the standardized indices SPI, SVI, and STI. The precipitation, NDVI, and temperature were fitted to the gamma distribution function through a mathematical transformation, whose probability density function (pdf) is expressed as

$$f(x) = \frac{1}{\beta^{\gamma}\Gamma(\gamma)} x^{\gamma-1} e^{-\frac{x}{\beta}}, \quad x > 0 \tag{7}$$

$$\Gamma(\gamma) = \int_0^{+\infty} t^{x-1} e^{-t} dt, \quad x > 0 \tag{8}$$

where x is the random variable of precipitation, NDVI, or temperature data, and β and γ represent the scale parameters and shape parameters of the gamma distribution, respectively, and can be fitted by maximum likelihood estimation. The cumulative distribution function (cdf) is calculated by integrating the pdf. Since the value conforms to the gamma distribution, which is greater than zero, the probability that the random variable x is less than an observed value x_i can be obtained as follows:

$$P(x < x_i) = \begin{cases} m/n, & x = 0 \\ \displaystyle\int_0^{x_i} f(x)\,dx, & x > 0. \end{cases} \tag{9}$$

After standardizing the probability value of the gamma distribution, the value of the inverse distribution function is obtained by the percentage point function (ppf) to calculate SPI, SVI, and STI.

B. Test for Goodness of Fit

To test the goodness of fit to establish whether SVI and STI are subject to the gamma distribution, we created a probability distribution object with parameters estimated from the time series and tested the null hypothesis that the data come from a population with a gamma distribution. If the value could reject the null hypothesis at the default 5% significance level, it can be regarded as conforming to the gamma distribution.

C. Calculation of SOSDI and Weight Matrix Analysis

SOSDI is the Hadamard product of three univariate indices and weight matrices, which are constructed by SPI, SVI, and STI. The expression of SOSDI is as follows:

$$\text{SOSDI} = \alpha \odot \text{SPI} + \beta \odot \text{SVI} + \gamma \odot \text{STI} \tag{10}$$

where α, β, and γ represent the weight matrix of SPI, SVI, and STI, respectively. $\odot$ represents the Hadamard product [58],

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TABLE I
LAND USE CLASSIFICATION IN THIS STUDY

Original type	Classification	Code
Cultivated land	Cultivated land	Farm
Forest	Forest and Shrubland	Fo_Sh
Grassland	Grassland	Grass
Shrubland	Forest and Shrubland	Fo_Sh
Wetland	Wetland and Water bodies	Wet
Water bodies	Wetland and Water bodies	Wet
Artificial surfaces	Bareland and Artificial surfaces	Ba_Built
Bareland	Bareland and Artificial surfaces	Ba_Built

TABLE II
CLIMATE CLASSIFICATION AND ITS FEATURES IN THIS STUDY

Features	Code
Tropical, rainforest	Af
Tropical, monsoon	Am
Tropical, savannah	Aw
Arid, desert	Bw
Arid, steppe	BS
Temperate, dry summer	CS
Temperate, dry winter	Cw
Temperate, no dry season	Cf

which means the multiplication of each element in the matrix and the value of the same position in another matrix. The constraint optimization method is introduced to calculate the weight matrix of SOSDI by treating SPEI and sc-PDSI as the objective functions. The weight matrix is represented as a 2-D arrangement of weights over geographical areas

$$f(x,y) = \max\left\{\frac{E[(X-ux)\times(Y-uy)]}{\sigma x \times \sigma y}\right\} \tag{11}$$

$$X = \begin{cases} \text{SPEI} \\ \text{scPDSI} \end{cases} \tag{12}$$

$$Y = \alpha \odot \text{SPI} + \beta \odot \text{SVI} + \gamma \odot \text{STI} \tag{13}$$

where $f(x,y)$ represents the function that makes X and Y reach the maximum correlation, E represents the mathematical expectation, ux and uy represent the average value of X and Y, respectively, σx and σy represent the standard deviation of X and Y, respectively, X represents the time series SPEI or sc-PDSI, and Y represents the traversal results for all combinations of standardized univariate variables, in which optimal matrices correspond to the weight of SOSDI. In the calculation, X and Y are the 1-D vectors that are converted from the original data. The weights α, β, and γ range from 0 to 1. Considering the accuracy and computation complexity, the parameter is changed with a step of 0.01 to optimize. Each weight follows the relations

$$\begin{cases} \alpha(i) = k \times 0.01, & k \in \mathbb{Z} \\ \beta(i) = j \times 0.01, & j \in \mathbb{Z} \\ \gamma(i) = 1 - \alpha(i) - \beta(i) \end{cases} \tag{14}$$

where k and j are the integers and are determined by the value of the weight coefficient. $\alpha(i), \beta(i)$, and $\gamma(i)$, respectively, represent the weight coefficients corresponding to pixel i. The range of $\alpha(i), \beta(i)$, and $\gamma(i)$ is [0.01, 0.98] since the minimum value of each weight is greater than zero.

The characteristics of the weight matrices were analyzed over various land cover and climate zones to highlight the primary factors determining the weight over geographical scenarios. The land cover was integrated into five categories: cultivated land, forest and shrubland, grassland, wetland and water bodies, and bareland and artificial surfaces. Climatic zoning takes into account the second-level classification system. The classification and their codes are shown in Tables I and II. According to the upper quartile Q3 and lower quartile Q1, the quartile distance IQR = Q3 − Q1 for each category was calculated. The upper and lower boundaries limit the normal value in the range of (Q1 − 1.5IQR, Q3 + 1.5IQR), beyond which ranges are defined as outlier data.

D. Classification Scheme of SOSDI

The classification of SOSDI can intuitively show the extent and evolution of drought. By calculating the cdf of SOSDI for a long time series, the classification scheme is determined after converting it into the percentile of the data distribution. Classification based on cumulative frequency percentage avoids the uncertainty caused by subjective division and reflects the fluctuation of historical drought. The cdf of SOSDI is expressed as follows:

$$F_X(x) = P(X \le x) \tag{15}$$

where $F_X(x)$ represents the probability that all values of SOSDI are no more than x. This probability is converted into a cumulative frequency percentile, and the percentile threshold is used as the demarcation of drought degree. The drought categories of SOSDI take the commonly used thresholds of 2%, 5%, 10%, 20%, and 30% [21], [59].

E. Performance Assessment

To explore whether the standardized drought index (SOSDI) had better performance, we focused on an in-depth comparison with the normalized drought index optimal scaled drought condition index (OSDCI) [20], which was composed of PCI, VCI, and TCI. Its expression was as follows:

$$\text{OSDCI} = \alpha \odot \text{PCI} + \beta \odot \text{VCI} + \gamma \odot \text{TCI} \tag{16}$$

where α, β, and γ represent the weight matrix of PCI, VCI, and TCI, respectively. Their calculations were referred to (11) and (12). The correlation was calculated to assess the performance of SOSDI and was expressed as

$$R = \frac{\sum\left(O_i - \bar{O}\right)\left(P_i - \bar{P}\right)}{\sqrt{\sum\left(O_i - \bar{O}\right)^2}\sqrt{\sum\left(P_i - \bar{P}\right)^2}} \tag{17}$$

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TABLE III
PERFORMANCE ASSESSMENT OF SOSDI BASED ON CROSS-VALIDATION OF DROUGHT INDICES

Objective function	Validation function	Proposed index	Compared indices
SPEI1	scPDSI	SOSDI1	MDSI1, DSI1, OSDCI1
scPDSI	SPEI1-6	SOSDI2	MDSI2, DSI2, OSDCI2

where O_i represents the values of SOSDI, P_i represents the values of evaluation indicators, and $\bar{O}$ and $\bar{P}$ represent their average values, respectively.

SOSDI was evaluated from several aspects. First, cross-validation was used to assess the performance of SOSDI with other drought indices (Table III). The results were compared with Z-scored DSI (drought severity index), MDSI (modified drought severity index), and normalized OSDCI.

Besides, soil moisture and crop yield were used to evaluate the SOSDI since they are effective indicators for reflecting the agricultural drought. The correlation between SOSDI and soil moisture was analyzed on a monthly scale, since the area is characterized by rainy and dry seasons with great seasonal variations. The relationship between annual SOSDI and crop yield was also analyzed.

Additionally, we conducted validation based on real drought events. According to the EM-DAT database by the Centre for Research on the Epidemiology of Disasters (CREDs) at the University of Leuven, severe drought events happened during 2005–2006 and 2010–2011 in East Africa. SOSDI was applied to analyze the characteristics of drought events in East Africa during typical periods of severe drought.

F. Sensitivity Analysis of Input Data

To test the sensitivity of input data, we perturb the original input data as an error term. The generated data with normal distribution noise can be expressed as

$$\text{Input}_{\text{error}} = \text{Input} \times (1 + \varepsilon \times \omega) \tag{18}$$

where $\text{Input}_{\text{error}}$ is the result with error terms, ε is a series of standard normally distributed random errors, and ω is the relative error levels ranging from 2% to 10% with an interval of 2%.

IV. RESULTS

A. Fitting Suitability of NDVI

MODIS NDVI was utilized as the main part and true values to calculate SVI since it is the most widely used and validated data, combined with the fitting GIMMS NDVI to meet the requirement of 30 years of records. The long-term trends of NDVI remained consistent before/after nearest resampling. The combined NDVI kept the same trend with MODIS NDVI, confirming the effectiveness of fitting and the rationality of fitting GIMMS NDVI to MODIS NDVI to calculate SVI.

The linear fitting result of all values from 2000 to 2015 showed the highest correlation and lowest deviation (Table IV and Fig. 3), indicating negligible NDVI saturation (logarithmic feature) and soil background effects (exponential feature) during fitting, since MODIS NDVI was believed to be reliable after quality control. The fitting results remained consistent with the long-term trend of MODIS NDVI. Pixel-by-pixel fitting was efficient, with R reaching 0.9727 ($p < 0.001$); PBIAS was 0.0004; RMSE was 0.0200. The results illustrated the efficiency of long time-series NDVI for in-depth analysis.

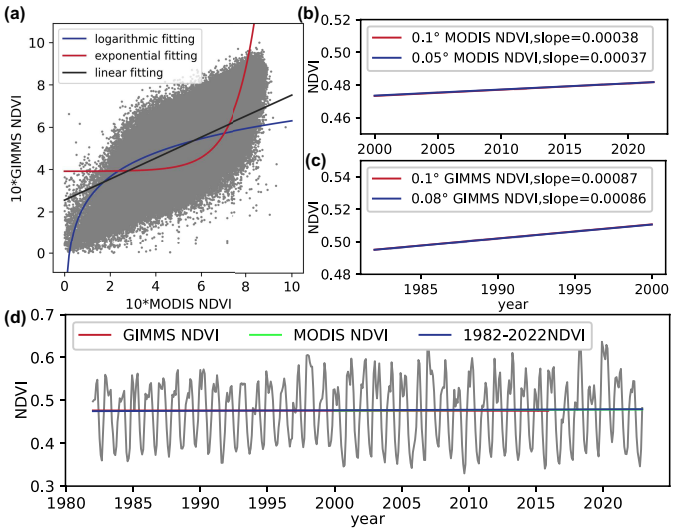


Fig. 3. Evaluation of fitting NDVI. (a) Comparison of linear, logarithmic, and exponential fitting results between GIMMS NDVI and MODIS NDVI. The long-term trends of (b) MODIS NDVI and (c) GIMMS NDVI before/after resampling. (d) Long-term trend of fitted NDVI.

TABLE IV
ESTIMATION MODEL OF GIMMS NDVI AND MODIS NDVI. THE x AND y REPRESENT MODIS NDVI AND GIMMS NDVI. THE VALUES OF NDVI HAD BEEN MULTIPLIED BY TEN

Model	Function	R	RMSE
logarithmic	$y = 1.641\ln(x) + 2.517$	0.932	1.170
exponential	$y = 0.002e^{x} + 3.916$	0.717	1.303
linear	$y = 0.496x + 2.551$	0.951	1.002

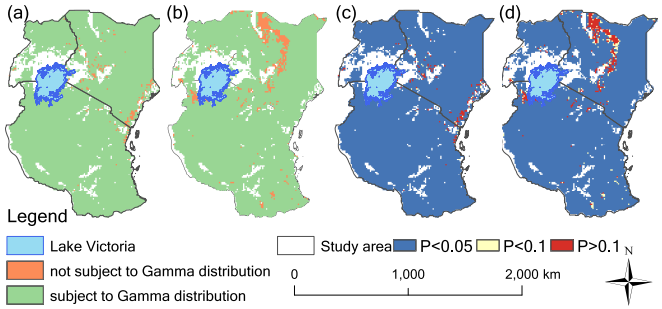


Fig. 4. Results of whether (a) SVI and (b) STI are subject to gamma distribution, and P values for (c) SVI and (d) STI.

B. Applicability of SVI and STI Fitting

The test for goodness of fit confirmed that SVI and STI were subject to the gamma distribution, which covered 98.43% and

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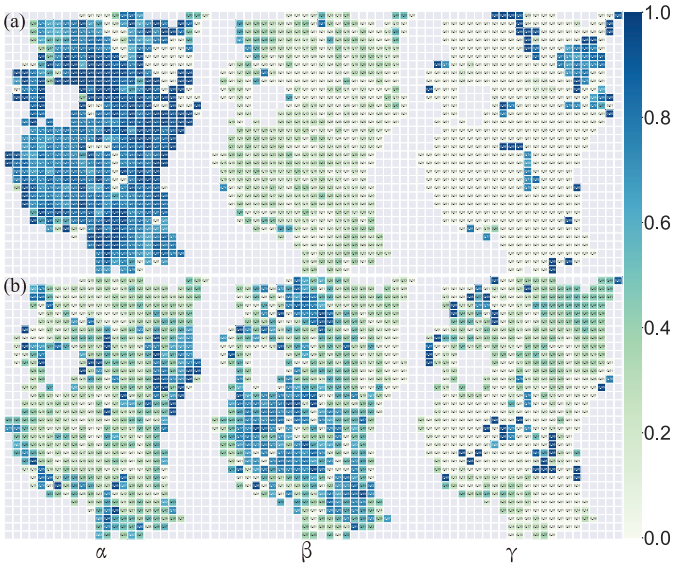


Fig. 5. Weight matrix of (a) SOSDI1 and (b) SOSDI2. α, β, and γ range from 0.01 to 0.98 and represent the weight of SPI, SVI, and STI, respectively.

95.54% areas (Fig. 4), proving the suitability of fitting NDVI and temperature to the gamma distribution. SVI and STI are reasonably developed and form the foundation of SOSDI.

C. SOSDI Weight Matrix and Its Feature

Weight matrices of SOSDI1 and SOSDI2 were constructed with SPEI and scPDSI as objective functions (Fig. 5), respectively. In the SOSDI1 weight matrix, the average weight of SPI was 0.71 and reflected greater impacts of precipitation. While the influence of SVI was moderate, reaching 0.16 on average, but higher than STI. The weight of STI was the lowest in most regions, except in the northeast area. This phenomenon illustrated that dominant impact of temperature only appeared in a few regions, where SPI had the lowest influence. In general, SPI impacted significantly on SOSDI1, SVI had a slightly lower impact, and STI only impacted in the northeast area.

The SOSDI2 weight matrix showed obvious geospatial heterogeneity. The average weight of SVI was generally higher and reached 0.43, reflecting the strong influence of the NDVI, especially in the area around Lake Victoria and southern Tanzania. The weight of SPI was second to SVI and reached its highest in the southeast corner of Kenya. The weight of STI was slightly higher than that of SPI and SVI in the northeast area and dominated only few areas in central Tanzania. Overall, SPI impacted moderately, SVI impacted prominently in the south, and STI had a relatively strong impact on the northeast.

The weight matrix of SOSDI exhibited distinct features over five land covers (Fig. 6). For SOSDI1, α was close in cultivated land, forest, and shrubland and grassland, whose weight in wetland and water bodies was higher, with an average value of 0.91. β was similar in most land covers, except for wetlands and water bodies, with a lower median value of 0.02. γ was high in bareland and artificial surfaces and fluctuated widely, with a median value of 0.15 and an average value of 0.43, while weights in other land covers were 0.01. For SOSDI2, α was close and stable, with the median floating around 0.3. β in wetland and water bodies was lower with a median value of 0.24 and an average value of 0.28, which was smaller than other land covers with corresponding values of 0.45 and 0.44. γ was high, and its median value was 0.31, which was higher than the median of other areas.

The weight matrix of SOSDI differed under climate zones. For SOSDI1, the mean value of α was similar and around 0.9, except for Bw zones. In the Bw zone, the weight fluctuated greatly, with a low average of 0.39 and a median of 0.18. β was lower with a median value of 0.01 in Af, Am, Cf, and CS zones, and a slightly higher median in arid climate types. The median value of the γ was 0.01 in almost all zones and was slightly higher in arid desert regions. For SOSDI2, the median value of α fluctuated around 0.3. The median and mean values of the β in Af and Bw zones were low, fluctuating around 0.25 and 0.32; the weight fluctuated greatly except for the CS zone, indicating the obvious effects of vegetation on SOSDI2. γ had more outliers over climate zones, which was three times that of α; and its median values in Aw and BS zones were significantly lower, being 0.01 and 0.06.

TABLE V
CRITERIA FOR DIVIDING SOSDI CATEGORIES

code	type	percentile k (%) of CDF
D4	Extreme drought	$k \leqslant 2$
D3	Severe drought	$2 < k \leqslant 5$
D2	Moderate drought	$5 < k \leqslant 10$
D1	Lightly drought	$10 < k \leqslant 20$
D0	Initial drought	$20 < k \leqslant 30$
N	Normal	$30 < k < 70$
W0	Initial wet	$70 \leqslant k < 80$
W1	Lightly wet	$80 \leqslant k < 90$
W2	Moderate wet	$90 \leqslant k < 95$
W3	Severe wet	$95 \leqslant k < 98$
W4	Extreme wet	$k \geqslant 98$

D. Classification of SOSDI

SOSDI was divided into 11 types based on the cumulative frequency percentile (Table V). According to Table VI, the threshold of SOSDI1 was lower than that of SOSDI2 when the cumulative frequency percentile was less than 64.26%. Using skewness and kurtosis to measure the symmetry of distribution and the peak of the highest frequency. The skewness of SOSDI1 was 0.4339, and the kurtosis was 0.5974, and the values were 0.2266 and 1.1838 for SOSDI2. SOSDI was mostly distributed to the left of the mean, while SOSDI2 had better symmetry and higher peak values than SOSDI1. The higher kurtosis reflected that SOSDI2 had more extreme values, which was consistent with the fluctuation of the weight distribution range.

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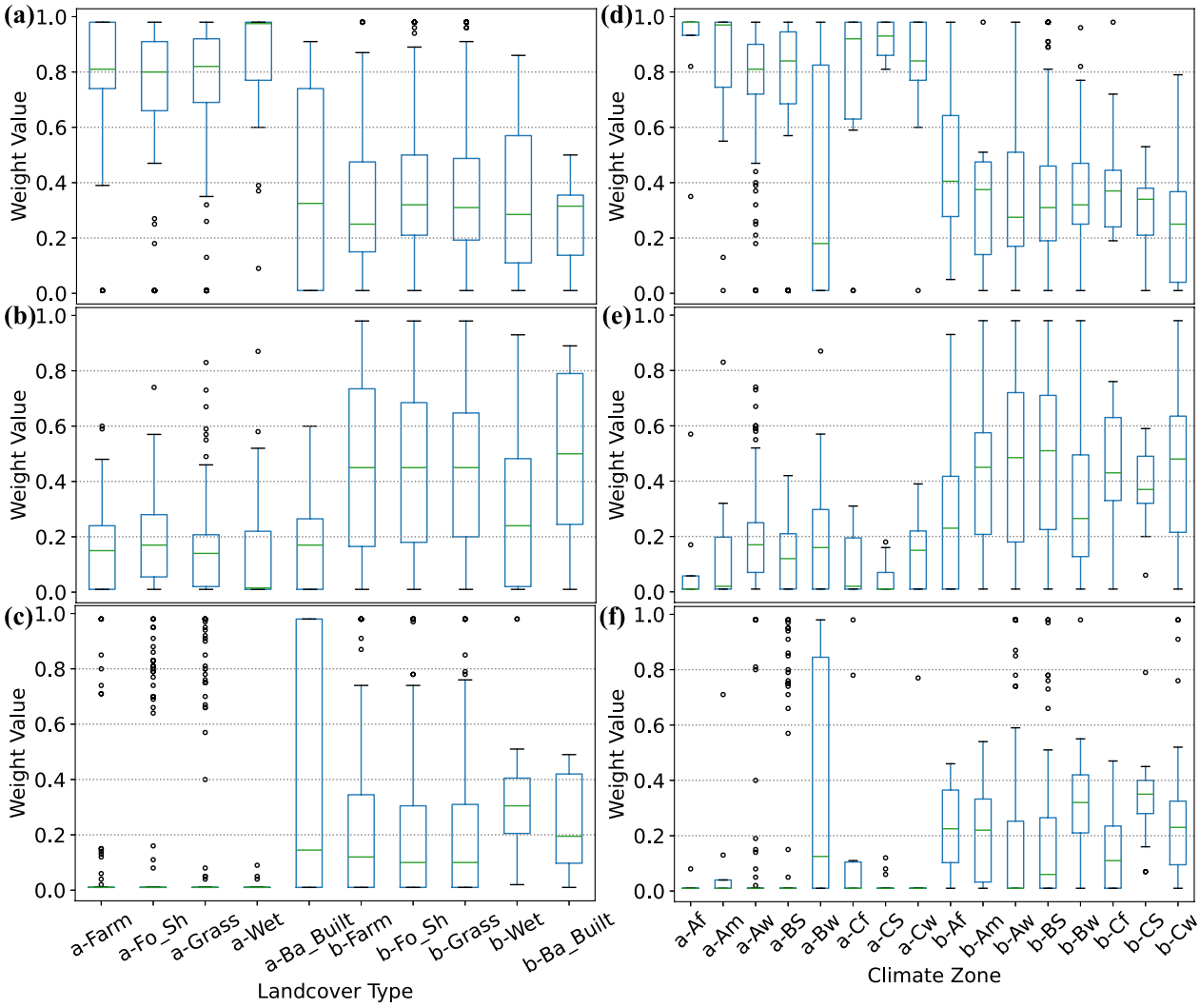


Fig. 6. (a)–(c) Boxplots of the weight matrix over land cover types and (d)–(f) climate zones. Weight of SPI in (a) and (d) weight of SVI in (b) and (e) weight of STI in (c) and (f) "a-" represents SOSDI1 and "b-" represents SOSDI2.

E. SOSDI Performance

1) The SOSDI was evaluated based on cross-validation of sc-PDSI and SPEI1-6. The standardized index was better than those by the normalized or Z-scored method (Fig. 7). The cross-validation eliminated the influences of objective functions. The results indicated that, whatever the combination of objective functions and validation, the standardized method and SOSDI achieved the best performance, far exceeding the normalization or Z-score method. From Fig. 7(b), as the SPEI time scale increased, the correlations with SOSDI2 did not decrease, while other indices showed decreases after SPEI3. This further highlighted the monitoring capability of SOSDI and its enormous potential for application.
2) The soil moisture fluctuated with alternating seasons. It was stable in the short-dry season from January to February, and increased with the precipitation and reached its highest in April, and slightly dropped in May. It decreased continuously due to the prolonged drought and showed the lowest level from July to September, which interfered with the validation. During the short-rainy season from October to December, the soil moisture gradually increased. The validation was

TABLE VI
K VALUES CORRESPOND TO THE SOSDI

k value	SOSDI1	SOSDI2
2	–1.5790	–1.4176
5	–1.2917	–1.0975
10	–1.0511	–0.8251
20	–0.7364	–0.5134
30	–0.4821	–0.2956
70	0.4317	0.3978
80	0.7327	0.6396
90	1.1578	0.9940
95	1.5376	1.3188
98	1.9987	1.7239

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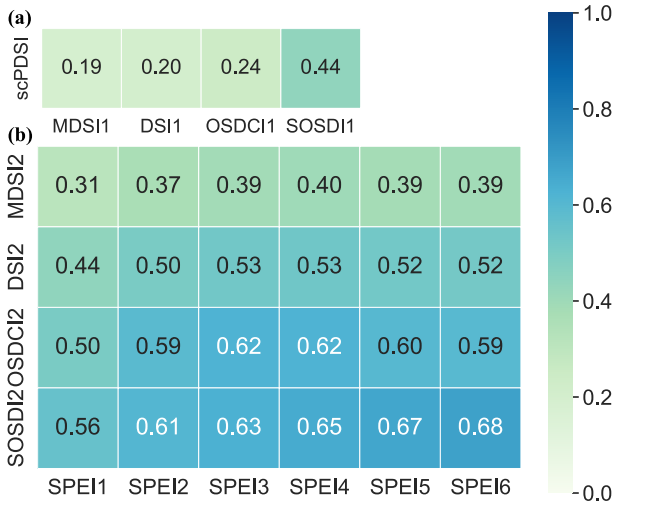


Fig. 7. Correlation cross-validation of SOSDI by (a) sc-PDSI and (b) SPEI ($p < 0.05$), compared with OSDCI, DSI, and MDSI.

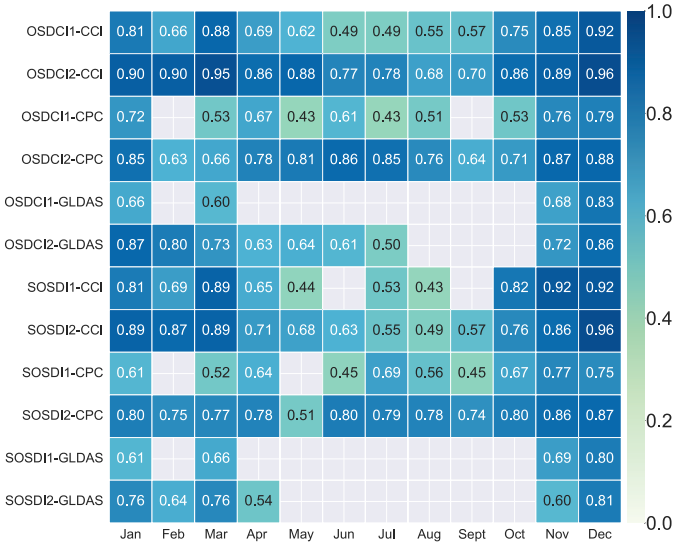


Fig. 8. Correlations of OSDCI, SOSDI, and soil moisture products (CCI, CPC, and GLDAS) at a monthly scale ($p < 0.05$).

on a monthly scale to reduce uncertainty (Fig. 8). The performance was related to the objective functions and differed over months. The correlations between SOSDI1 and soil moisture were generally better than OSDCI. However, SOSDI2 showed less advantage with soil moisture. SOSDI could better reflect the changes in soil moisture and perform similar to OSDCI. The correlations between SOSDI and soil moisture were in order from high to low: short-rainy season, short-dry season, long-rainy season, and long-dry season. The lack of water in the long-dry season lowered the soil moisture, presenting a weak correlation. The soil water content was saturated in the long-rainy season, exceeding the average under normal conditions, with a correlation slightly lower than that of the short-rainy and short-dry seasons under unsaturated conditions. The recovery of soil moisture in the short-rainy season was highly consistent with the gradual relief of drought and positively influenced the correlation.

3) Both SOSDI and OSDCI showed good correlations with crop yield, and SOSDI performed better (Fig. 9). The data distributions based on SPEI1 and sc-PDSI were more similar after standardization and provide a comparative perspective for exploring the impact of the objective functions. However, the index obtained by the normalization method was quite different, especially in the data ranges. Under detrending analysis, the correlations between SOSDI1 and crop indicators were significantly better than OSDCI1.
4) SOSDI could monitor real drought events (Figs. 10 and 11). It captured the severest drought in October–December 2005 and well monitored the drought in southern Tanzania and northern Kenya during 2010–2011, consistent with other studies [9], [60]. It could capture the signs of drought, its development, and evolution, as well as the process of eventual regression. This demonstrated the effectiveness of SOSDI during true events.

F. Sensitivity of SOSDI

Sensitivity results indicated that SOSDI had good resistance to disturbances in input data (Fig. 12). This is related to the methods. The mapping of data positions rather than numerical values ensures the robustness of weights.

V. Discussion

A. Advantages of the Standardized Method

The effective information extracted by data normalization and standardization varies greatly. Normalization only applies to stable data and is sensitive to extreme and outlier values. By mapping the location of data, standardization is more suitable and enables the SOSDI constructed with objective functions to be comparable, which is the strength of a standardized approach [17], but previously only applied to univariate drought indices (e.g., SPI and SPEI). SOSDI extends standardization to the synthesized drought index and realizes better monitoring of agricultural drought. Compared with OSDCI, SOSDI correlates better with the validated indicators. SOSDI also has the stability to monitor prolonged drought. Existing soil moisture products are highly uncertain [61], and spatial models may not reflect the real situation [62], let alone the lack of effective verification in Africa. It suggests considering standardized methods in the assimilation system and retrieval algorithm of soil moisture.

The standardized approach highlights the influence of temperature with more drought information (Fig. 13). Although SOSDI was mostly positively correlated with sc-PDSI and SPEI1, it was still negatively correlated in a large range. SOSDI decreased with the increase of sc-PDSI and SPEI1 in the negatively correlated area, which was caused by rising temperature. The negatively correlated area showed a high degree of coincidence with the area that showed higher values of γ. The coincidence rate of SOSDI1-SPEI1 and its higher weight pixels was 100%, and the average weight of STI

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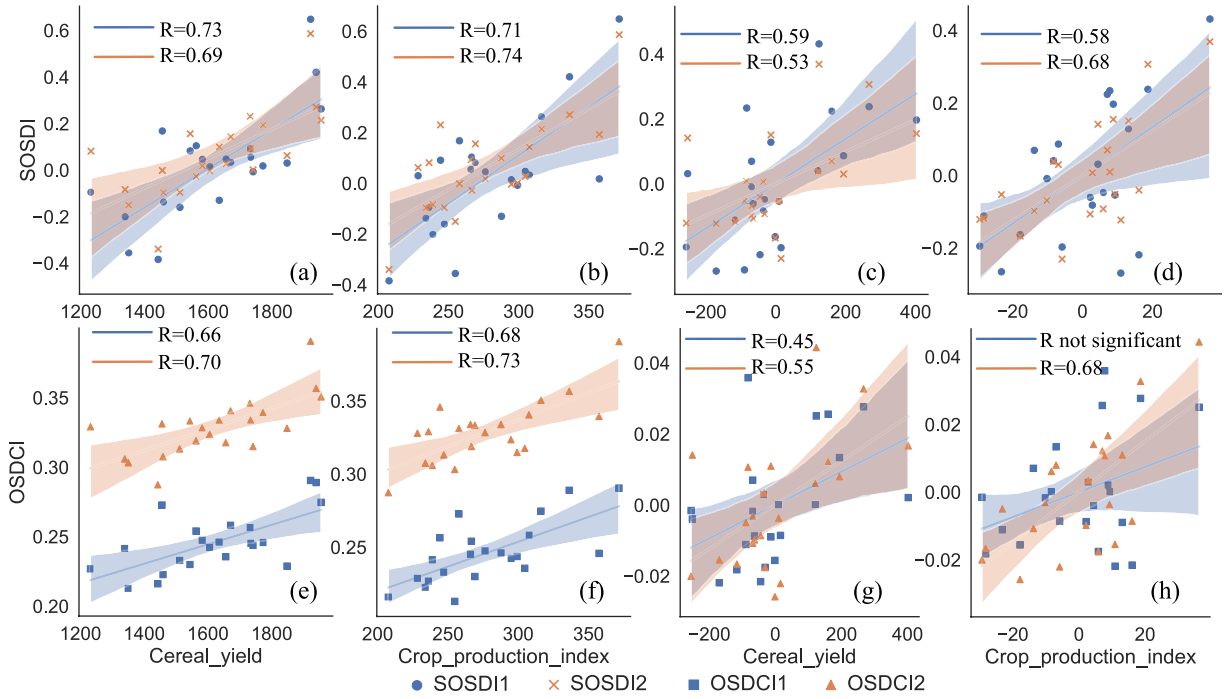


Fig. 9. (a), (b), (e), and (f) Correlations between drought indices (SOSDI and OSDCI) and crop indicators. (c), (d), (g), and (h) Detrended correlations between drought indices (SOSDI and OSDCI) and crop indicators ($p < 0.05$).

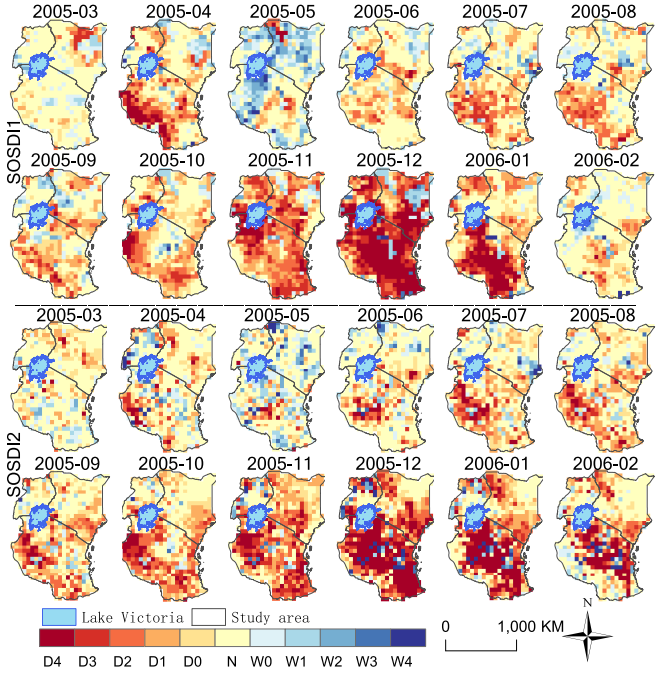


Fig. 10. Spatial and temporal features of the drought event during 2005–2006 in the area.

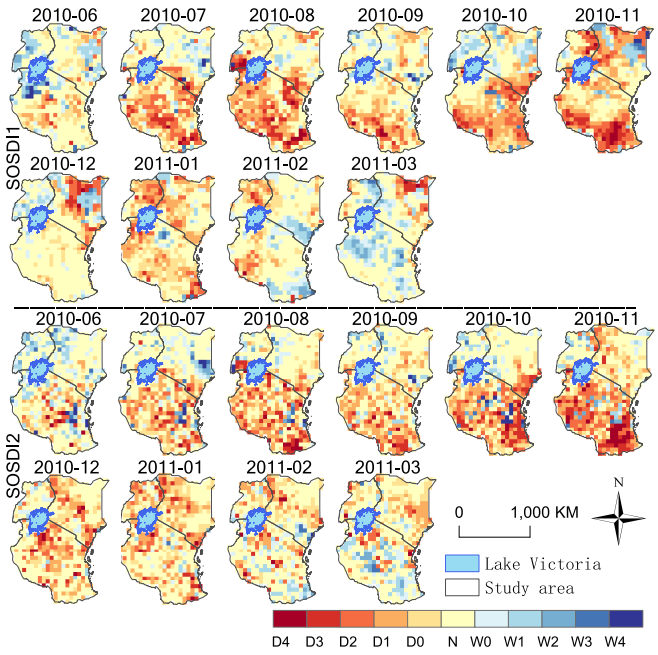


Fig. 11. Spatial and temporal features of the drought event during 2010–2011 in the area.

was 0.87. The coincidence rate of SOSDI2-sc-PDSI with the weight greater than 0.5 was 85%. That is what OSDCI cannot detect.

The effective expression of vegetation information can be demonstrated by the high weight of vegetation in the proposed weight matrix. The average weight of SVI in SOSDI1 reached 0.16 and exceeded the impact of temperature. The average weight of SVI in SOSDI2 is the highest, reaching 0.43, and fully expresses its impact. SVI expresses more drought-related information and illustrates the strengths of standardization.

B. SOSDI Weights Over Geographical Environments

The weight matrix is connected to and greatly impacted by the geographical environment. We primarily investigate the effects of land cover and climate zones, offering a reference for

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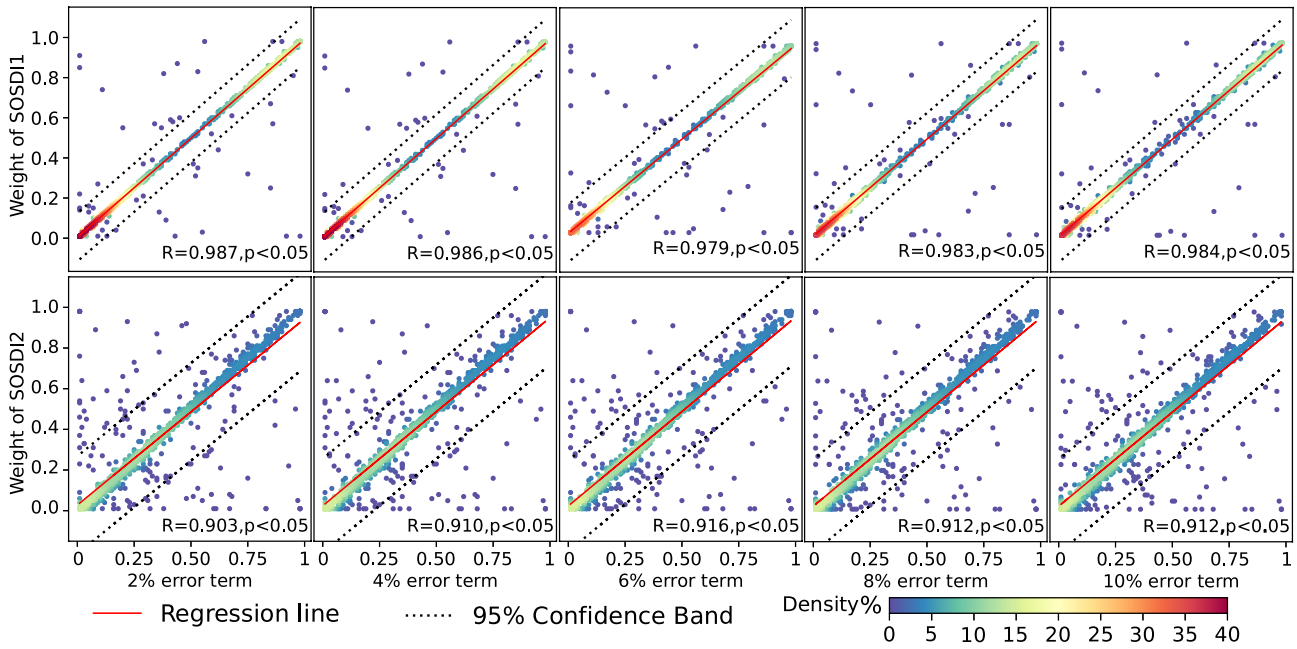


Fig. 12. Sensitivity of SOSDI under standard normally distributed random error, with relative levels of terms from 2% to 10%.

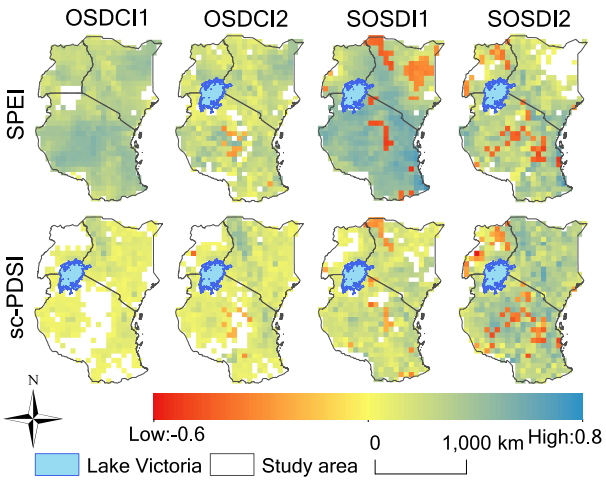


Fig. 13. Spatial correlation between OSDCI, SOSDI, and sc-PDSI, SPEI1 ($p < 0.05$).

understanding the mechanisms of drought and their linkages with the surface environment.

Over land covers, α of SOSDI1 was generally higher than that of SOSDI2, while β and γ of SOSDI1 were generally lower than those of SOSDI2. This was determined by the objective functions. Precipitation was the dominant factor in SPEI, resulting in higher α, lower β, and γ of SOSDI1. α of SOSDI1 was low over the Ba_Built area, while γ was high. β of SOSDI2 was lower over the wet area, while γ was higher. This reflected that the weight of SOSDI1 was more sensitive to bareland and artificial surfaces, while that of SOSDI2 was more sensitive to wetlands and water bodies. β of SOSDI2 showed high values over farm, forest and shrubland, and grass, confirming the effect of vegetation.

α of SOSDI1 was generally higher than that of SOSDI2, while β and γ were lower. γ was higher over the Bw zone with fluctuation, covering the minimum to the maximum values, reflecting the instability of the desert climate region. The weight of SOSDI1 was close, and the weight of SOSDI2 was more affected by the climate and was unstable.

The distinct performance of the weight matrix reminds us to pay attention to the background and formation of drought. Although we emphasize that the scope is agricultural drought monitoring instead of investigating drought formation, it is still important to take more environmental factors into account.

C. Drought Features Indicated by SOSDI in East Africa

1) The interannual variation of SOSDI with different severities from 2000 to 2021 is shown in Fig. 14. Severe drought was observed around 2000, 2005, and 2011. According to the categories of SOSDI1, the proportion of D4 and D3 reached the highest of 5.77% and 6.21% in 2005. SOSDI2 reflected the most severe drought in 2000, with D4 and D3 accounting for 4.97% and 7.11%. The proportions of wet pixels reached highest and exceeded 40% during 2018–2020. The wettest year was 2020 and 51.6% of pixels were classified as wet, with the highest W4 and W3 ratios of 7.35% and 7.66% in SOSDI1. SOSDI2 was the wettest in 2020, and the proportion was 61.6%, with W4 and W3 accounting for 9.63% and 10.29%.
2) From a seasonal perspect, the average value of SOSDI1 was greater than 0 in the rainy seasons and less than 0 in the dry seasons and reached the lowest at −0.078 (Fig. 15). The outliers reached 1.89% in the long-dry season and 1.12% in the short-dry season, while did not exceed 0.80% in rainy seasons. The mean values of SOSDI2 were all greater than 0, with the highest

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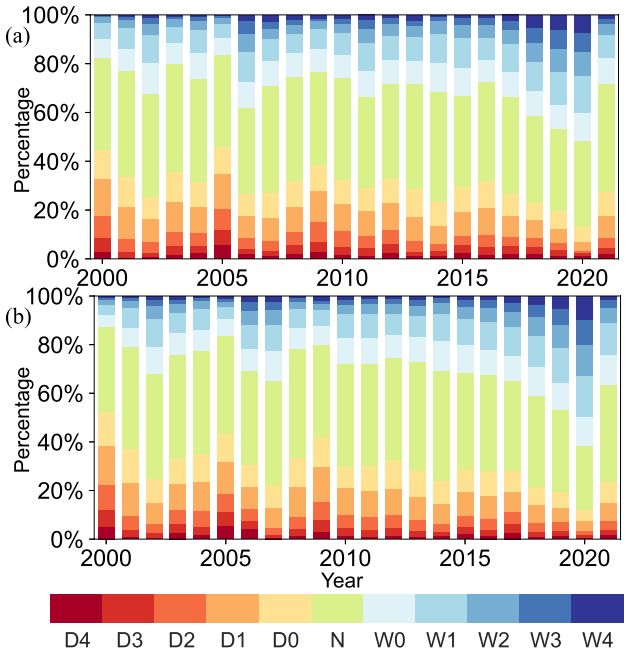


Fig. 14. Annual classifications of drought degree illustrated by (a) SOSDI1 and (b) SOSDI2 from 2000 to 2021.

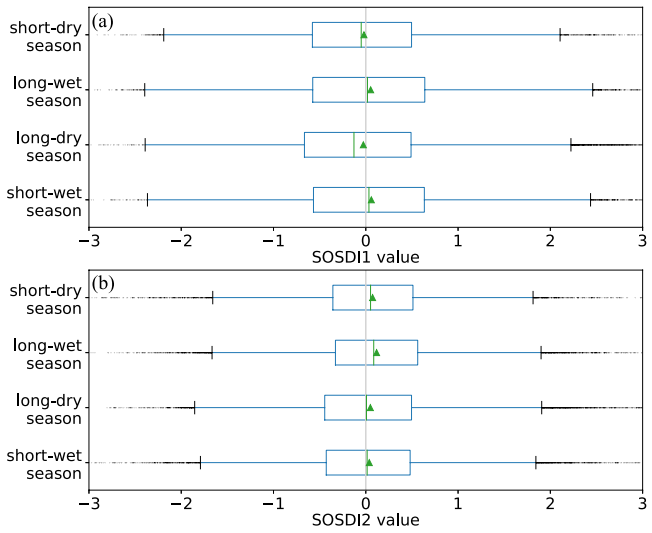


Fig. 15. Drought features of (a) SOSDI1 and (b) SOSDI2 at seasonal scale by box plot in the same seasons. Green lines and triangles represent the median and mean values, respectively.

value of 0.110 in the long-rainy season and the lowest value of 0.021 in the long-dry season. The distribution of SOSDI2 was characterized by high average, low maximum, and high minimum values, reflecting narrow data ranges. The outliers of SOSDI2 were high, ranging from 2% to 3%. In general, SOSDI1 could indicate drought more intuitively based on the mean value with fewer data anomalies, which was more consistent with seasonal characteristics.

D. Robustness of SOSDI

To test the robustness of SOSDI, the cross-validation of Table III in another important agricultural area, South Africa,

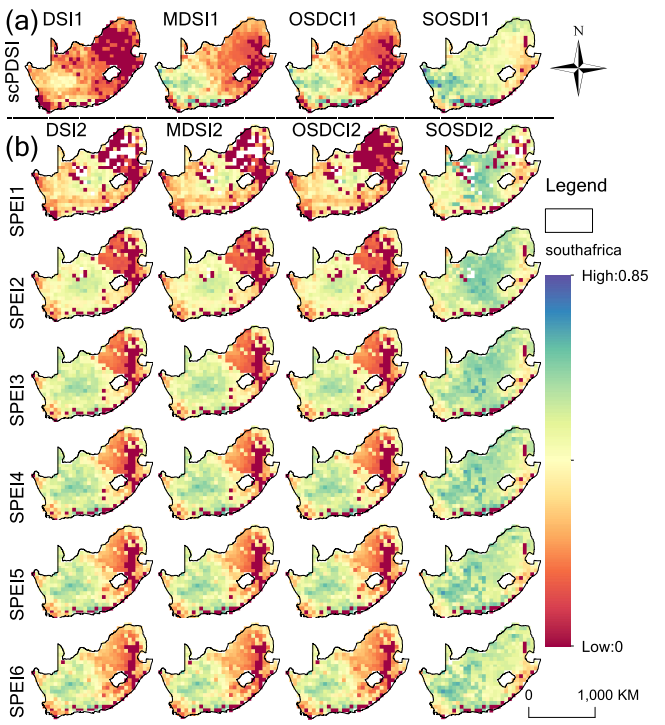


Fig. 16. Absolute correlation of cross validation by (a) sc-PDSI and (b) SPEI in South Africa, $p < 0.05$. Comparison including SOSDI, OSDCI, DSI, and MDSI.

was conducted. SOSDI showed the highest correlations in the cross-validation (Fig. 16), reflecting its robustness.

E. Influence of the Objective Function

As a reference for constraint optimization, the objective functions determine the performance of SOSDI and whether one factor can take effect. The initial OSDCI and other indices only used SPEI as a reference and lacked a comparison of various objective functions [20], [63], [64]. The former indices suppress vegetation and temperature according to the weight matrix of SOSDI1, and it is well improved by SOSDI2. It is necessary to compare the influence of objective functions.

SPEI1 is more dependent on precipitation, and sc-PDSI has more impacts on vegetation and temperature. During the assessment, the correlation of SOSDI1 and SOSDI2 decreases with the increase of the SPEI time scale. This is due to the delayed response of vegetation to drought, which has been widely verified [65], [66], and can only be reflected in SOSDI2, where vegetation occupies greater proportions. SOSDI2 correlated better with soil moisture since vegetation responds to soil conditions better than precipitation. For the correlation with crop yield, SOSDI1 is significantly better than OSDCI1, while SOSDI2 is less improved than OSDCI2. Because vegetation occupies a large proportion in both SOSDI2 and OSDCI2, the similar information reflected by vegetation causes a consistent correlation with crop yield indicators.

The objective functions have great impacts and need to be considered for better comparison. The weight matrix proposed in this study is an effective tool to intuitively represent and

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directly detect the impact of the objective functions on SOSDI, which is also applicable to other drought indices.

F. Uncertainty of Soil Moisture on Validation

Due to the uncertainty in soil moisture product retrieval, we considered its impact and took measures to reduce the influence of the uncertainties on the validation.

First, the covering effect of vegetation will reduce the signal and exacerbate the underestimation of deep soil moisture, which phenomenon can also be caused by sensor limitations with insufficient penetration depth into the soil. Thus, the surface soil moisture data were used for validation to overcome the high uncertainty of retrieval in the deep soil moisture.

Besides, the drought might not be well captured and cause seasonal differences, considering the vegetation cover varied greatly in the dry/wet seasons. The underestimation or overestimation of soil moisture varies significantly in different seasons, which uncertainty will cause inaccurate results if the soil moisture of all months is compared together, considering their inherent overestimation and underestimation are different. Thus, the validation is on a monthly scale separately to address this irregular bias.

At last, there are other uncertainties in soil moisture retrieval that are difficult to quantify, including the lack of validation from ground stations. Our solution and advantage lie in using multiple independent soil moisture products for validation, which greatly reduces the impact of single soil moisture product uncertainty. The results indicate that this strategy is effective, as SOSDI generally demonstrates good performance. We strongly suggest that validation in drought research requires the use of multiple independent soil moisture products to mitigate the uncertainty impact on individual products.

G. Fitting Suitability of the Gamma Distribution

To maintain consistency in the standardization and compare the effects of different inputs and explore their spatial differences, it is necessary to use the same distribution to provide comparability for weight matrices. This is the reason that precipitation, NDVI, and temperature were fitted to the same distribution. From the expression of the gamma function, the shape parameter α and scale parameter β can flexibly adapt to the skewness and kurtosis of different variables. This means that it can capture the skewness of variables such as NDVI by changing the two parameters.

The gamma distribution has been fitted to precipitation [67] and has been widely used to calculate SPI [68], [69], [70], serving as the most common method for drought monitoring. The fitting of NDVI and temperature to the gamma distribution originates from the fact that variables with linear relationships are subject to the same distribution function with different parameters. The linear correlations between precipitation and NDVI and temperature have been confirmed in East Africa [71], [72], [73]. Thus, it is reasonable to fit NDVI and temperature to the gamma distribution, the same as the precipitation.

This adaptability is also supported by experimental results. The test for goodness of fit already confirmed that the gamma distribution was suitable to calculate SVI and STI (Fig. 4).

Although the results already confirm the validity of fitting suitability of the gamma distribution, it is helpful to compare more distributions based on geographic locations in the future, which will help expand the application of standardization.

H. Limitations and Future Work

This work aims to promote the application of standardization and has great potential in drought monitoring. Still, there are some aspects that can be improved.

Although the NDVI used in the study had good accuracy and low error, it is suggested to conduct NDVI fusion research and field validations, such as Bayesian and machine learning methods, when promoting its global application. Besides, the MODIS NDVI was thought to be reliable and represents true values in the study, and the results indicate that it had already met the research requirements, and using MODIS NDVI as the main data is highly necessary for many regions around the world that lack on-site observations, such as East Africa. Still, it is valuable to investigate the saturation and soil background effects of NDVI [74], [75] with on-site surveys and compare with field tests for accurate correction in the future.

Due to the limitation of input data resolution, the resolution of SOSDI is not high. Downscaling is the solution, but it requires observations, which are scarce in East Africa. To avoid introducing errors without in situ validation, we did not increase the resolution. However, this does not affect the application of SOSDI and can be addressed partly by the weight matrices. Higher spatial resolution data are also applicable to calculate SOSDI, which can be applied to monitor different scales of drought. Besides, the features of the weight matrices in Fig. 6 are still valid for high-resolution inputs. It is applicable to monitoring drought based on SOSDI and comparing drought features at different spatial resolutions in the future.

VI. Conclusion

Based on SPI, SVI, and STI, SOSDI was developed to fill the gap in the absence of a standardized synthesized drought index. The proposed weight matrix was found to be crucial and could reflect the sensitivity of factors to land covers and climate types, making it a useful tool for understanding what constitutes drought monitoring. Standardization performed better in the synthesized drought index compared to normalization or Z-score methods, which was confirmed by the multidimensional evaluations of SOSDI. It could reflect drought information of vegetation and temperature better by enhancing their expression and improving the drought monitoring effect.

Severe drought in East Africa was captured around 2000, 2005, and 2011, mainly from June to September. The average value of SOSDI1 and SOSDI2 reached their lowest in the long-dry season, which were −0.078 and 0.021.

The development of SOSDI is critical for agricultural drought monitoring and vulnerability assessment in Africa and provides support by assessing drought-prone areas for early disaster prevention and reduction. Its applications include predry season water storage and allocation in vulnerable

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regions, crop pattern optimization based on the features of weight matrices to adapt to climate factors, integrating interpretability of weight matrices into the drought monitoring system, and spatial yield mapping based on the relation between crop yield and SOSDI.

ACKNOWLEDGMENT

The authors appreciate the following data providers and website maintenance organizations. The CHIRPS data were available at https://data.chc.ucsb.edu/products/CHIRPS-2.0/. Global Inventory Monitoring and Modeling Study (GIMMS) Normalized Difference Vegetation Index (NDVI) was available at https://ecocast.arc.nasa.gov/data/pub/gimms/, and MODIS NDVI was available in the MODIS repository https://search.earthdata.nasa.gov/search. The LST was retrieved from ERA5 at https://www.ecmwf.int/en/era5-land. The sc-PDSI was retrieved from https://crudata.uea.ac.uk/cru/data/hrg/, and Standardized Precipitation Evapotranspiration Index (SPEI) was retrieved from https://spei.csic.es/database.html. Soil moisture provided by ESA Climate Change Initiative (CCI) was available at https://climate.esa.int/en/, from Global Land Data Assimilation System (GLDAS) was available at https://disk.gsfc.nasa.gov/datasets/, and from Climate Prediction Center (CPC) was available at https://psl.noaa.gov/data/gridded/data.cpcsoil.html. Crop datasets from World Bank were available at https://data.worldbank.org.cn. The geographic background data of land cover were retrieved from GlobeLand30 at https://www.webmap.cn/commres.do?method=globeIndex

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